Fiscal Year:	FY 2019	**Task Last Updated:**	FY 04/08/2020
PI Name:	Hyers, Robert Ph.D.		
Project Title:	Modeling and Simulation of Electrostatically Levitated Multiphase Liquid Drops		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rwhyers@wpi.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	508-831-5759
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City:	Worcester	**State:**	MA
Zip Code:	01609-2280	**Congressional District:**	2
Comments:	Formerly with University of Massachusetts at Amherst.		
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Japanese Space Agency (JAXA) AO for International Research Participation on ISS Kibo Module
Start Date:	04/08/2014	**End Date:**	07/31/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Craven, Paul	**Contact Phone:**	256-544-7649
Contact Email:	pcravenRRR@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 7/31/2019 per NSSC information (Ed., 5/10/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lee, Jonghyun Ph.D. (Iowa State University) Watanabe, Masahito Ph.D. (JAXA PI: Gakushuin University, Japan)		
Grant/Contract No.:	NNX14AH85G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Project continues with "Modeling and Simulation of Electrostatically Levitated Multiphase Liquid Drops--80NSSC20K1038," start date 5/1/2020 and same Principal Investigator, Dr. Robert Hyers. Funding is for Dr. Hyers's role as U.S. Principal Investigator (PI) on Professor Masahito Watanabe's (Department of Physics, Gakushuin University, Tokyo, Japan) proposal, “Interfacial Phenomena and Thermal Properties of High-Temperature Melts,” funded by the Japanese Aerospace Exploration Agency (JAXA). Prof. Masahito Watanabe's project: During steel making process, such as continuous casting, the impurity in the cast steel is influenced by the interplay between the molten steel and molten slags. Understanding the interfacial phenomena is necessary to produce high-purity steels. An accurate measurement of interfacial tension between molten steel and molten slag is one essential step to understand the interfacial phenomena. Well-controlled oscillation of a sample with a concentric core (molten steel) – shell (molten slag) structure is required to measure the interfacial tension. Due to the effect of gravity, this concentric arrangement cannot be obtained using ground-based facilities. An Electrostatic Levitation Furnace (ELF) in the KIBO deck in the International Space Station (ISS) will be utilized for the measurement of interfacial tension at the interface between molten steel and molten slag. For the success of the space mission, it is of critical importance to predict the interfacial tension and the influence of main parameters, such as the core-shell ratio, thermophysical properties of molten slags, magnitude of initial deformation, magnitude of electrostatic forces, and off or inphase oscillation, on the predicted interfacial tension. With such information, the space experiment can be designed efficiently and effectively. Dr. Hyers's (University of Massachusetts) scope of work as part of Dr. Watanabe’s proposal is summarized below: 1. Simulation of the flow in electrostatically levitated core-shell droplets, including dynamics such as oscillations and viscous effects on the dynamics. 2. Simulation of the effect of changes in the dimensions of the core and shell on the surface oscillations. 3. Simulation of the time dependence of the changes in surface oscillations due to reactions between the oxide and steel phases. 4. Design, direct, and analyze experiments to be conducted in the Marshall Space Flight Center (MSFC) Electrostatic Levitation Facility. The Research is to use data from an Electrostatic Levitator (ESL) facility supplied by the Japanese to be used on the International Space Station, and also a ground based ESL facility located at MSFC.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of the project are expected to be utilized for more efficient production of high-purity steels, which are the main metals used on Earth.
Task Progress:	FINAL REPORTING SUBMITTED APRIL 2020 Numerical Modeling • Numerical modeling of an oscillating droplet with core-shell structure by Aribtrary Lagrangian-Eulerian method in OpenFOAM. A single phase droplet oscillation was successfully modeled with the interface tracking method using OpenFOAM. • A single phase droplet oscillation was successfully modeled with the volume of fluid method using ANSYS Fluent. • First flight results were achieved by our partners in Japan in 2018. • A dual-phase droplet oscillation was successfully modeled with the phase-field method. Project Team Meetings and Team Results • Attended the first meeting with the ELF project collaborators, Nagoya, Japan, September 2014. • Face-to-face meeting with JAXA PI Prof. Watanabe in 2015 to discuss latest experimental results from parabolic flight campaign. A novel phenomenon was observed, where a metal-slag droplet did not form the usual core-shell structure. Instead, the droplet formed a double droplet. From the images, Prof. Watanabe’s group was able to determine the interfacial energy between the metal and slag phases. UMass is modeling this case and its dynamics. • Measured interfacial tension between Fe and welding flux / Fe and BF slag in ELF on Parabolic Flights in May 2016. • Additional flight results are being analyzed with the new models. • Additional flight experiments are presently planned through at least calendar year 2022 (by Prof. Watanabe). ANNUAL REPORTING SUBMITTED MARCH 2019: • A dual-phase droplet oscillation was successfully modeled with the phase-field method. • Presented results at the 148th The Minerals, Metals & Materials Society (TMS) Annual Meeting in San Antonio, TX, March 2019. ED. NOTE: Project continues with "Modeling and Simulation of Electrostatically Levitated Multiphase Liquid Drops--80NSSC20K1038," start date 5/1/2020 and same Principal Investigator, Dr. Robert Hyers. See that project for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 03/30/2026)

Abstracts for Journals and Proceedings	Lee J, Sumaria K, Rabeh A, Ganapathysubramanian B, Hyers RW. "Numerical Simulations of Oscillation of Single and Multi-Phase Metal Droplets." 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Oct-2018
Abstracts for Journals and Proceedings	Lee J, Hyers RW, Matson DM, Rogers JR, Katamreddy S, Jo YJ, Lee GW. "Containerless Materials Processing:Thermophysical Properties and Solidification of Molten Metals and Aqueous Solutions." TMS 2019. 148th Annual Meeting, The Minerals, Metals and Materials Society, San Antonio, TX, March 10-14, 2019. TMS 2019. 148th Annual Meeting, The Minerals, Metals and Materials Society, San Antonio, TX, March 10-14, 2019. , Mar-2019
Abstracts for Journals and Proceedings	Rabeh A, Khanwale M, Ganapathysubramian B, Hyers R, Lee J. "Computational Fluid Dynamics Modeling of Oscillation Damping of Compound Liquid Droplets." TMS 2020. 149th Annual Meeting, The Minerals, Metals and Materials Society, San Diego, CA, February 23-27, 2020. TMS 2020. 149th Annual Meeting, The Minerals, Metals and Materials Society, San Diego, CA, February 23-27, 2020. , Feb-2020
Articles in Peer-reviewed Journals	Sumaria K, Hyers RW, Lee J. "Numerical prediction of oscillation behaviors of a multiphase core-shell droplet during interfacial tension measurement." Metallurgical and Materials Transactions B. 2019 Dec;50(6):3012-9. Published online 11 September 2019. https://doi.org/10.1007/s11663-019-01680-3 , Dec-2019